Lab 1: Project Requirements

Objective

The objective of this lab is to understand the project requirements and how it helps in satisfying the client's needs. Hence, ensure successful project outcomes.

What is a requirement?

A requirement is something the product must do or a quality it must have. Mastering the Requirements Process.

The requirement should describe:

- **What** a system shall do (function).
- **How well** the system shall do its functions (system quality or performance level).
- A resource or design **limitation** (constraint or budget).

Types of Requirements:

- **User requirements:** Statements in natural language plus diagrams of the services the system provides and its operational constraints. It is written for customers.
- **System requirements**: A structured document setting out detailed descriptions of the system's functions, services and operational constraints. It is written as a contract between client and contractor.

Example 1:

User Requirements definition: The software must provide means of representing and accessing external files created by other tools.

System Requirements specification:

1. The user should be provided with facilities to define the type of external files.
2. Each external file type may have an associated tool which may be applied to the file.
3. Each external file type may be represented as a specific icon on the user's display.
4. Facilities should be provided for the icon representing an external file to be defined by the user.
5. When a user selects an icon representing an external file, the effect of that selection is to apply the tool associated with the type of external file to the file represented by the selected icon.

Example 2: The Child Care Management software

User Requirements definition: When children arrive in the morning the teaching assistant will record the time of their arrival. For example, when the children leave, the assistant will record the time of their departure. The departure time will be used to keep track of late pick-ups so the parent can be billed accordingly.

System Requirements specification:

1. Only the teachers and assistants shall be allowed to enter or edit child comments and child arrival or departure times.
2. Each user shall be required to log in with a unique username and password before using the system.
3. Parents will be charged 10 $ per hour, per child, for late pick-ups.

Functional and Non-Functional Requirements

- Functional Requirements: What should the system do?
  - Functional requirements describe what the system should do.
  - Statements of services that the system should provide, how the system should react to inputs and how the system should behave in specific situations.
  - Functional requirements will probably trace to specific functions of a program.
- Non-functional Requirements: What must the system adhere to?
  - Non-functional requirements are global constraints on a software system.
  - Constraints on the services or functions offered by the system such as timing constraints, constraints on the development process, standards, development costs, operational costs, performance, reliability, maintainability, portability, robustness etc.
  - Usually, cannot be implemented in a single module of a program.
  - Represents the quality of requirements.

Examples on Functional Requirements

Example 1:

- The user shall have the option to search across all the initial databases or select a specific subset to search.

Example 2: The Child Care Management software

- If only a single child is enrolled a parent shall be billed $90 at the end of each month.
- If only two children are enrolled a parent shall be billed $157.50 at the end of each month.
- If more than two children are enrolled, parents shall be billed $157.50 for the first two children, plus $45 for each additional child, at the end of each month.
- Each classroom shall have a maximum capacity of 20 children.
- Each student shall only belong to one class.
- One teacher and one assistant shall be assigned to each class.
- The administrators shall be able to enter the type and schedule of required immunizations.
- A notification regarding required immunizations shall be included on the monthly

invoice.

- The immunization due and history report shall list children alphabetically by last name, detailing any required immunizations and the dates of previous immunizations.

Examples on Non-Functional Requirements

Use **shall** for mandatory requirements.
Use **should** for optional requirements.

Property	Measure
Speed	Processed transactions/s User/Event response time Screen refresh time
Size	Kbytes Number of RAM chips
Ease of Use	Training time Number of help frames
Reliability	Mean time to failure Probability of unavailability Rate of failure occurrence Availability
Robustness	% events causing failure Time to restart after failure Probability of data corruption on failure
Portability	% target system dependent statements Number of target systems

Figure 1 Sample measures of non-functional requirements

1) Usability:

Describes how easy it is for the intended users of the product to operate it. For example, minimizing the training time it takes from users to comfortably use the application.

Examples

- The product shall use symbols and language that are intuitive and easily understood by the user community.
- The product shall be able to be used by members of the public with no prior training.
- The product shall be used by engineers who will attend five weeks of training before using the product.
- Users must be able to begin using the system within 15 minutes or less.
- The application should always inform the user of time-consuming operations by displaying a message on the screen.
- The WinZip program shall show a progress bar during the file compression or decompression process.

2) Reliability and availability:

- Mean Time Between Failures (MTBF) tells how much time (in hours, weeks, or months) must pass between system failures. On the other hand, Mean Time to Repair (MTTR) specifies how long the system is allowed to be out of use (that is, unrepaired) after a failure. An example of reliability requirement is: "The System service X shall have an availability of 999/1000 or 99%" which means that out of every 1000 requests for this service, 999 must be satisfied.

Examples

- The ATM system shall have a maximum failure rate of one occurrence every two months.
- The product shall be available for use 24 hours per day, 365 days per year.
- The product shall be available for use during the hours of 8:00 am to 5:30 pm.
- The Mean Time Between Failures (MTBF) shall vary by use case, e.g., over two weeks for a website but less than 15 minutes for an elevator.
- The product shall achieve 99% up time.
- During system repair, a backup server shall be activated to continue delivering the requested service.

3) Capacity:

Specifies the volumes that the product must be able to deal with and the amount of data stored by the product. To ensure that the product can process the expected volumes.

Examples

- The product shall be designed to support up to 300 simultaneous users between 9:00 A.M. and 11:00 A.M. During other times, the maximum load will be 150 simultaneous users.

4) Performance:

Specifies the amount of time available to complete specified tasks. These requirements often refer to response times. They can also refer to the product's ability to operate at a speed suitable for the intended environment.

Some products—usually real-time products—must be able to perform some of their functionality within a given time slot. Failure to do so may mean catastrophic failure (e.g., a ground-sensing radar in an airplane fails to detect an upcoming mountain) or the product will not cope with the required volume of use (e.g., an automated ticket-selling machine).

Examples

- System Y shall process a minimum of 8 transactions per second.
- Any user interface in the automated system shall have a maximum response time of 2 seconds.
- The product shall poll the sensor every 10 seconds.
- All web pages must load within three seconds under average load conditions, and within five seconds during peak load.

- The maximum response time for an office application on a LAN should be 1 second.
- The application must start in less than two seconds.
- The web application must be capable of handling 100 transactions per second.

5) Security:

Refers to the ability to prevent and/or forbid access to the system by unauthorized parties.

Confidentiality is the requirement that data and processes be protected from unauthorized disclosure.

Integrity is the requirement that data and processes be protected from unauthorized modification.

Examples

- Only authorized payroll administrators shall be permitted to access employee pay information.
- The access permissions for system data may only be changed by the system's data administrator.
- Only direct managers can see the personnel records of their staff.
- To protect system files from malicious users or programs (such as viruses and worms), antivirus/antispyware software should be installed on the file server and regularly updated.
- Only the system's data administrator is authorized to modify access permissions for system data.
- All system data must be backed up every 24 hours, with backup copies stored in a secure location separate from the system's physical location.
- All external communications between the system's data server and clients must be encrypted.

6) Safety

Defined as the absence of catastrophic consequences on the environment.

Examples

- The nuclear reactor control system shall shut down the reactor when its temperature exceeds the maximum allowable limit.
- The elevator/garage door shall not close when it detects an object on the door's path.
- The system shall not operate if the external temperature is below 4 degrees Celsius.
- The system should no longer operate in case of fire (e.g. an elevator).

7) Design constraints:

They impose restrictions on how the product must be designed. For example, it might have to be implemented in the hand-held device being given to major customers, or it might have to use the existing servers and desktop computers, or any other hardware, or software.

Examples

- The size (dimensions) of the hardware housing the system shall not exceed (10 cm height x 8 cm width x 1cm thickness).

Attributes of Requirements

- **Unambiguous:** It has a clear and single interpretation.
- **Verifiable:** Is the ability to test and demonstrate that a requirement has been met or not.
- **Traceable:** Is the ability to track the relationship between requirements and other project artifacts throughout the development process.
- **Abstract:** It defines ***what*** needs to be done, not ***how*** it should be done.

Recommendations When Writing Requirements

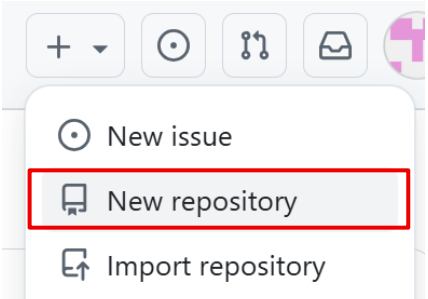
- **Never assume others understand your intentions**: Always provide clear and explicit information.
- Be clear and concise. Avoid using ambiguous language that may confuse stakeholders.
- Be specific and provide enough detail for developers to understand exactly what needs to be built.
- Prioritize requirements based on their importance to the project.
- Validate requirements with stakeholders to ensure that they are accurate and complete.
- **Use precise, meaningful words**: Avoid vague terms like "process," "manage," "perform," "handle," and "support."
- Focus on requirements, not features:
  - **Feature:** Broad and tested only for existence.
  - **Requirement:** Specific, measurable, and testable.
- Avoid the following:
  - **Conjunctions**: Consider whether the requirement should be divided into two distinct requirements.
  - **Conditionals**: Refrain from using terms like "if," "else," "but," "except," or "although."
  - **Possibilities**: Avoid words like "may," "might," "probably," or "usually."
  - **Subjective or imprecise terms**: Stay clear of vague words like "quickly," "easy," "timely," "before," "after," "user-friendly," "effective," "multiple," "appropriate," "normal," "capability," "reliable," "state-of-the-art," or "effortless."

Lab 1 Work:

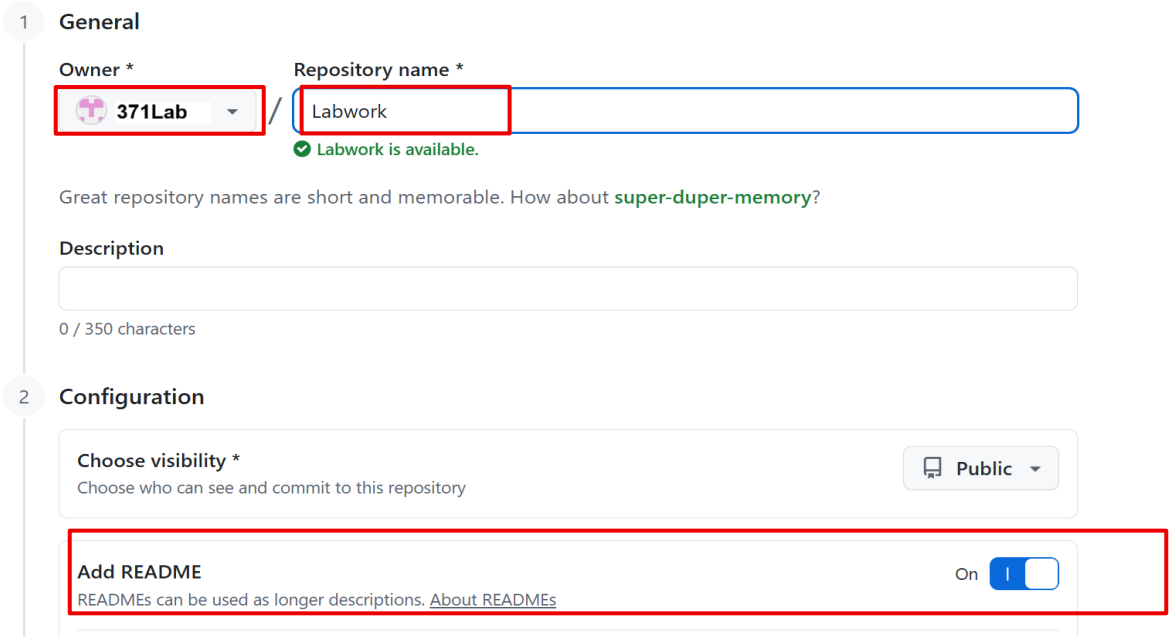
- Mention 1 or 2 Functional requirements and 1 or 2 non-functional requirements of the Student Registration System.
- Write two functional requirements and two non-functional requirements of online store. Discuss the requirements with other students and exchange comments, then ask the engineer to review them.

Steps to upload PDF files to GitHub using the GitHub UI

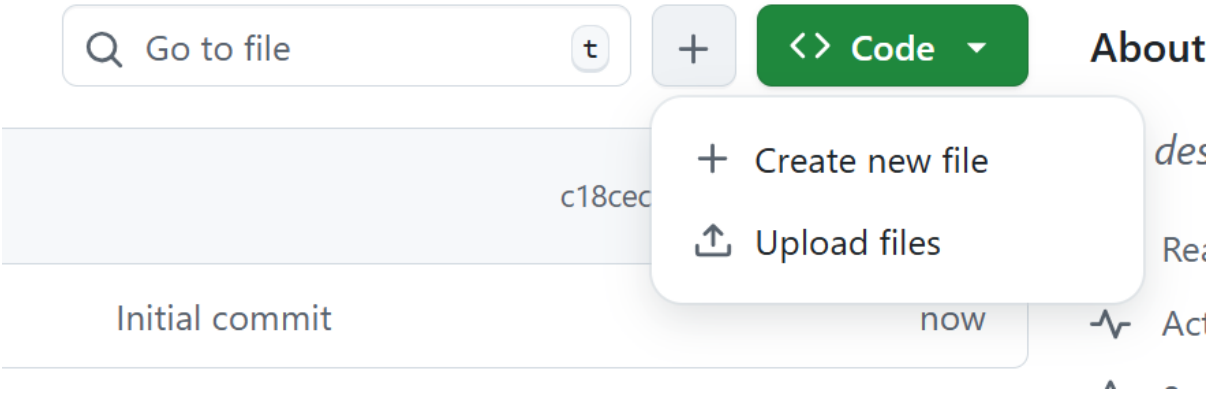
1. Log in to your GitHub account at github.com
2. Click the plus sign (+) and select **"New repository".**



3. Select the owner, enter a repository name, and initialize with a README file. Finally, click the **"Create repository"** button.



4. Click the plus (+) button, then choose **"Upload files"**.



5. Drag and drop files from your computer into the upload area or click **"choose your files"** to select them manually.

Drag files here to add them to your repository

Or choose your files